NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83401/21
13 August 1991

PERFORMANCE SPECIFICATION

RESISTOR NETWORK, FIXED, FILM,
STYLE RZ210, 6 PIN SIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RZ210, 6 pin single-in-line package (SIP) resistor networks. This style is available in all characteristics and resistance tolerances.

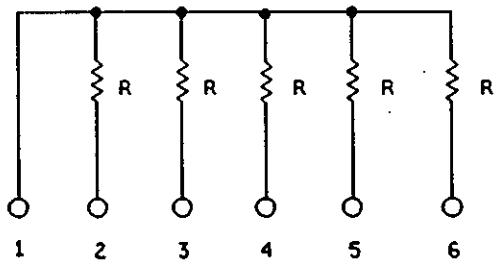
1.2 Military part number. Resistor networks covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:

M8340121 — Detail specification number

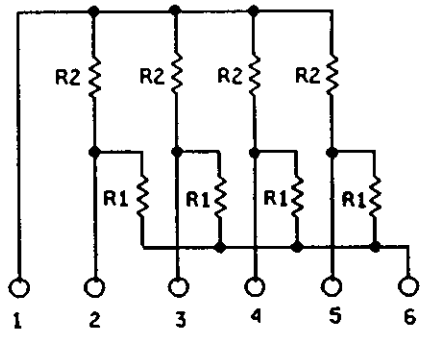
H1002FC — Coded number

The coded number shall be derived in accordance with MIL-R-83401.

1.2.1 Schematics. The schematic of the resistor network shall be identified by a single letter in accordance with the drawings below. The resistor element R_{REF} shall be the reference element used in determining the ratio accuracy (when applicable).



SCHEMATIC C



SCHEMATIC H

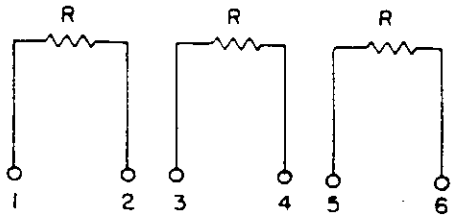
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5905

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SCHEMATIC G

1.2.1.1 Standard resistance values. The standard resistance values and the resistance value designators for the H schematic are as specified in table I.

TABLE I. Standard resistance values.

Resistance designator	R_1 (ohms)	R_2 (ohms)	Resistance designator	R_1 (ohms)	R_2 (ohms)
A001	82	130	A010	330	470
A002	120	200	A011	330	680
A003	130	210	A012	1.5k	3.3k
A004	160	260	A013	3.0k	6.2k
A005	180	240	A014	180	270
A006	180	390	A015	270	270
A007	220	270	A016	560	560
A008	220	330	A017	560	1.2k
A009	330	390	A018	620	2.7k

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-R-83401 - Resistor Networks, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Resistors, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-83401, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Internal connection. Internal connections shall be accomplished by thermal compression bonding of gold to gold, to form a solid phase weld from the lead to the internal resistor connection. No solder connection, eutetic or noneutetic, shall be permitted in this construction.

3.3 Power rating (70°C). The power ratings for schematics C, G, and H shall be as specified in table II.

TABLE II. Power ratings.

Schematic	Characteristic	Element power rating (watts)	Network power rating (watts)
C	C	0.06	0.3
	V	0.06	0.3
	H	0.12	0.6
	K	0.12	0.6
	M	0.12	0.6
G	C	0.06	0.18
	V	0.06	0.18
	H	0.12	0.36
	K	0.12	0.36
	M	0.12	0.36
H	C	0.04	0.3
	V	0.04	0.3
	H	0.07	0.6
	K	0.07	0.6
	M	0.07	0.6

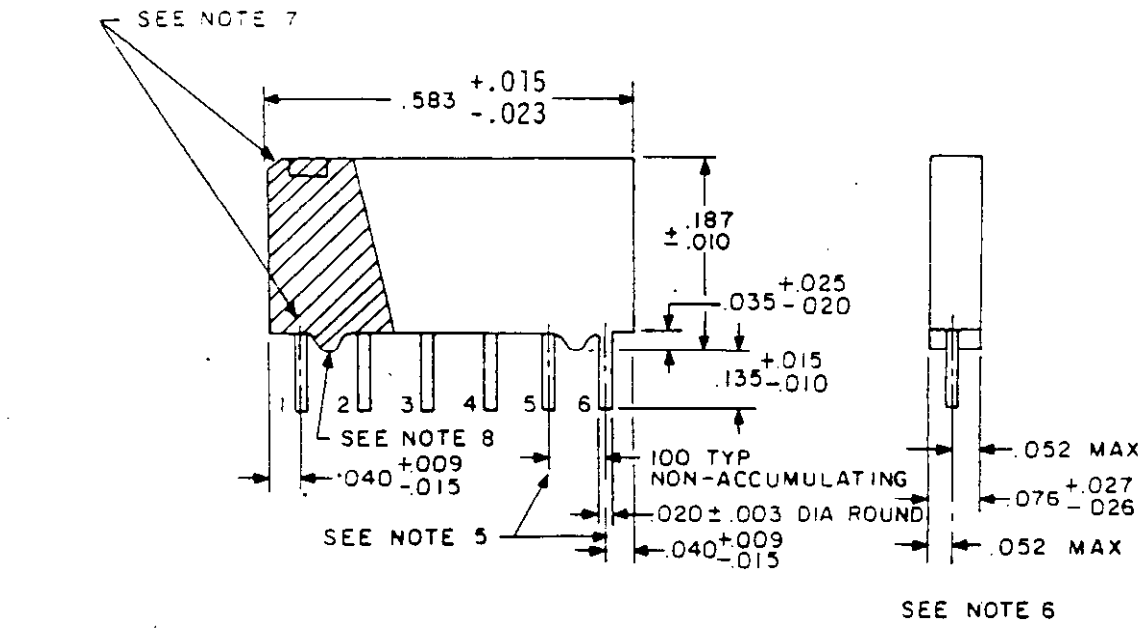
3.4 Power conditioning. The power applied for power conditioning for schematics C, G, and H shall be 1.5 times rated power.

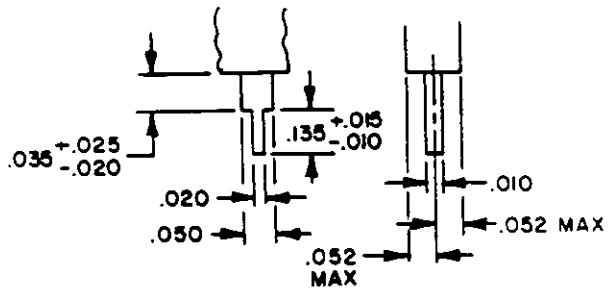
3.5 Voltage rating. The maximum continuous working voltage for each resistor shall not exceed 50 volts dc or ac rms.

3.6 Resistance. Minimum and maximum resistance values shall be as follows:

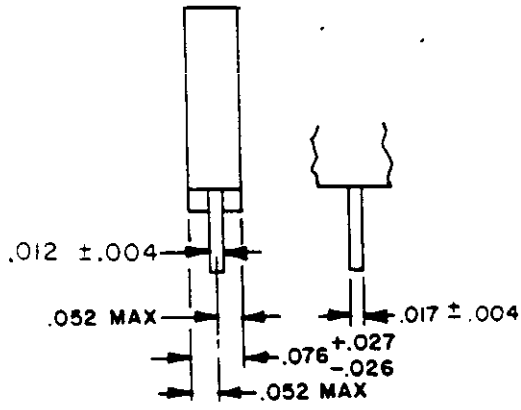
Resistance tolerance	Minimum resistance (ohms)	Maximum resistance (megohm)
B (±0.1 percent)	100	0.1
D (±0.5 percent)	100	0.1
F (±1.0 percent)	10.0	1
G (±2.0 percent)	10.0	1
J (±5.0 percent)	10.0	1

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Optional lead configuration with standoff as part of lead (see note 8)



Optional lead configuration with standoff as part of body (see note 8)

FIGURE 1. Style RZ210 resistors.

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Inches	mm	Inches	mm
.003	0.08	.026	0.66
.004	0.10	.027	0.69
.005	0.13	.030	0.76
.009	0.23	.035	0.89
.010	0.25	.040	1.02
.012	0.30	.052	1.32
.014	0.36	.076	1.93
.015	0.38	.089	2.26
.017	0.43	.100	2.54
.020	0.51	.135	3.43
.023	0.58	.342	8.69
.025	0.64	.983	24.97

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. The picturization of the style above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.
5. Terminal centerline to centerline measurements made at point of emergence of the lead from the body.
6. Measurement made at point of emergence of the lead from the body.
7. Pin 1 locator shall be a dot, notch, stripe, or numeral 1 adjacent to pin no. 1, in the shaded area. Also, additional marking may be placed on the top edge where the bevel may be located.
8. If the standoffs are located on the body, a minimum of two standoffs are required as illustrated. As an option, additional standoffs may be located on the body of the resistor network. If leads with standoffs are used, standoffs on the body are not required.

FIGURE 1. Style RZ210 resistors - Continued.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-83401.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-83401 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5905-1270-7)

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1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-83401/21	

3. DOCUMENT TITLE
Resistor Network, Fixed, Film, Style RZ210, 6 Pin SIP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 98.29(

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